

Primitive Affine Local-Obstruction Splitting at a Fixed Determinant:

Squarefree Support, Exact Root Counts, and the Remaining Liouville
Parity Correlation

Liang Wang

School of Artificial Intelligence and Automation

Huazhong University of Science and Technology, Wuhan 430074, P.R. China

wangliang.f@gmail.com

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Abstract

Let

$$L_1(r) = d + sr, \quad L_2(r) = u + ar, \quad su - ad = h_0 \neq 0,$$

with $\gcd(d, s) = \gcd(u, a) = 1$. We give an exact local-obstruction audit for this primitive affine pair. The two determinant identities

$$\gcd(a, s) \mid h_0, \quad \gcd(L_1(r), L_2(r)) \mid h_0$$

show that all simultaneous value divisors are confined to the fixed prime set dividing h_0 .

For every prime p and $k \geq 1$, let $\nu_{p,k}$ be the number of residues modulo p^k on which $p^k \mid L_1(r)$ or $p^k \mid L_2(r)$, and set

$$\epsilon_{1,p} = \mathbf{1}_{p \nmid s}, \quad \epsilon_{2,p} = \mathbf{1}_{p \nmid a}.$$

We prove the exact formula

$$\boxed{\nu_{p,k} = \epsilon_{1,p} + \epsilon_{2,p} - \epsilon_{1,p}\epsilon_{2,p}\mathbf{1}_{p^k \mid h_0}.$$

Thus a common root modulo p^k exists exactly when both slopes are units modulo p and $p^k \mid h_0$. In particular, the simultaneous-squarefree local factor is

$$1 - \frac{\nu_{p,2}}{p^2} > 0,$$

and

$$\mathfrak{S}_{\text{sf}}(L_1, L_2) := \prod_p \left(1 - \frac{\nu_{p,2}}{p^2}\right) \geq \prod_p \left(1 - \frac{2}{p^2}\right) > 0.$$

We prove that this Euler product is the natural simultaneous squarefree density for every fixed primitive pair. We also give the exact square-divisor expansion: for squarefree q_1, q_2 , the system

$$q_1^2 \mid L_1(r), \quad q_2^2 \mid L_2(r)$$

is soluble precisely when

$$(q_1, s) = (q_2, a) = 1, \quad \gcd(q_1, q_2)^2 \mid h_0,$$

and then defines one residue class modulo $\text{lcm}(q_1^2, q_2^2)$.

On every interval on which both forms are positive,

$$\mu(L_1(r))\mu(L_2(r)) = \underbrace{\mu^2(L_1(r))\mu^2(L_2(r))}_{\text{squarefree/local support}} \underbrace{\lambda(L_1(r))\lambda(L_2(r))}_{\text{Liouville parity}}.$$

This separates all square-power local obstructions from the remaining two-sign problem without solving the latter. The unresolved arithmetic target is uniform cancellation for the literal growing fixed- h_0 weighted correlation, with its actual origins, masks, prefixes, polarizations, and outer reassembly. The algebra, root counts, Euler product, and divisor expansion are L0. Their attachment to the complete physical packet gives a conditional L1 certificate schema whose literal verification remains open. The required growing-family Liouville cancellation is L2 and is not proved. The determinant compatibility ledger $\lambda_D \leq 2\eta_Z$ and the independent physical-loss rule $\Lambda_{\text{phys}} < 1/400$ remain separate. No prime-pair lower bound, parity breach, or twin-prime conclusion is claimed.

Keywords: affine forms; fixed determinant; local obstruction; squarefree support; Liouville correlation; parity barrier; fixed shift.

Literal-frame convention. Every physical statement retains one prescribed nonzero determinant h_0 , the literal growing affine coefficients, the actual interval origin, all native masks and weights, both polarizations, every outer key, and one global post-aggregation normalization. No statement is specialized to $h_0 = 2$.

1 Introduction

Consider two integral affine forms

$$L_1(r) = d + sr, \quad L_2(r) = u + ar \quad (1)$$

whose coefficient determinant is prescribed:

$$su - ad = h_0 \neq 0. \quad (2)$$

This is the literal algebraic shape returned by the primitive-column gate of the fixed-shift TPC route [5]. The same determinant appears in the two-point logarithmic Chowla setting, but the physical packet has growing coefficients, ordinary weighted fibers, prescribed origins, masks, and a nontrivial outer reassembly. Those quantifiers cannot be removed by notation.

The present paper asks a deliberately narrower question:

Which part of the two-Mobius coefficient is completely explained by primitive affine local arithmetic, and which part remains a genuine parity correlation?

The answer is exact. The fixed determinant confines every common value divisor to h_0 . Primitivity then gives a closed formula for the root count modulo every prime power. At the squarefree level, the forbidden residues form an explicit periodic local support with a positive Euler product. None of these facts cancels the remaining Liouville signs.

1.1 The three proof levels

We use the proof-level convention of the surrounding TPC series.

- L0: finite integer algebra, exact congruence counts, Chinese-remainder identities, fixed-form density theorems, and structure-free logical separations;
- L1: verification that the exact objects are the literal physical coefficients on their actual intervals, with every mask, origin, weight, polarization, key, and global normalization retained;
- L2: a new uniform arithmetic cancellation estimate for the resulting growing fixed- h_0 family.

This paper proves L0 theorems and formulates an exact L1 crosswalk. It proves no L2 estimate.

1.2 Main results

Assume first that the forms are primitive:

$$\gcd(d, s) = \gcd(u, a) = 1.$$

For a prime p and $k \geq 1$, write

$$\mathcal{R}_{p,k} = \{r \bmod p^k : p^k \mid L_1(r)\} \cup \{r \bmod p^k : p^k \mid L_2(r)\}, \quad \nu_{p,k} = \#\mathcal{R}_{p,k}.$$

The exact count is

$$\nu_{p,k} = \mathbf{1}_{p \nmid s} + \mathbf{1}_{p \nmid a} - \mathbf{1}_{p \nmid as} \mathbf{1}_{p^k \mid h_0}. \quad (3)$$

On the two-nonconstant branch $as \neq 0$, this formula distinguishes two kinds of exceptional prime:

- (i) a *joint* or collision prime divides h_0 , hence belongs to a fixed finite set when h_0 is fixed;
- (ii) a one-slope prime divides exactly one of a, s , may move with the growing coefficients, and deletes one local root rather than creating a simultaneous obstruction.

In particular, a collision modulo p requires $p \mid h_0$, while a collision modulo p^2 requires the strictly stronger condition $p^2 \mid h_0$.

The squarefree support is

$$\mathcal{Q}(r) = \mu^2(|L_1(r)|) \mu^2(|L_2(r)|),$$

away from the at most two zeros of the forms. Its local density is

$$\mathfrak{S}_{\text{sf}}(L_1, L_2) = \prod_p \left(1 - \frac{\nu_{p,2}}{p^2}\right) > 0.$$

For fixed coefficients this is the natural density of $\mathcal{Q}(r) = 1$. This is a special linear instance of the classical squarefree-pattern framework of Mirsky [2]; we include a complete elementary proof because the exact dependence on the slopes and h_0 matters here.

Finally, on a positive active interval,

$$\mu(L_1)\mu(L_2) = \mathcal{Q} \lambda(L_1)\lambda(L_2).$$

The local factor $\mathcal{Q}$ is fully classified. The surviving weighted correlation of the two Liouville signs is not. Tao's two-point theorem treats fixed affine forms with logarithmic averaging [3]; averaged-shift theorems change a different quantifier [1]. Neither theorem supplies the literal growing-coefficient, fixed- h_0 , ordinary weighted, uniform-prefix statement required here; the logarithmic-to-ordinary gap is audited in TPC-87 [8].

1.3 What the split does not say

A positive squarefree Euler product says that no finite collection of square-power congruences annihilates the support. It does not give cancellation of $\lambda(L_1)\lambda(L_2)$, does not select a prescribed shift from an averaged theorem, and does not produce a lower bound for primes. In a growing physical family, even the passage from a formal local product to uniform support mass on every short translated fiber requires its own estimate.

The conclusion is therefore a precise fork. A positive continuation must prove sign cancellation on the literal squarefree support. A negative continuation may show that a proposed theorem cannot retain the growing forms, interval origins, or endpoint budget. Either outcome must keep the compatibility and physical-loss ledgers independent.

1.4 Organization

Section 2 proves the determinant and content identities. Section 3 gives the complete prime-power root classification. The squarefree Euler product, fixed-form density, and uniformity warning appear in section 4. Section 5 proves the exact Mobius–Liouville and square-divisor decompositions. Section 6 identifies the unresolved literal correlation. The independent ledgers and final claim boundary are recorded in sections 7 and 8.

2 Determinant algebra and primitive reduction

Throughout, divisibility is in $\mathbb{Z}$ and every greatest common divisor is positive.

Definition 2.1 (Primitive affine pair). The pair (L_1, L_2) in (1) is *individually primitive* if

$$\gcd(d, s) = 1, \quad \gcd(u, a) = 1.$$

No coprimality between the two slopes is assumed.

Theorem 2.2 (Fixed-determinant gcd confinement). *Let L_1, L_2 be as in (1) and suppose (2). Then*

$$\boxed{\gcd(a, s) \mid h_0}, \tag{4}$$

$$\boxed{\gcd(L_1(r), L_2(r)) \mid h_0 \quad (r \in \mathbb{Z})}. \tag{5}$$

More generally, if an integer m divides both $L_1(r)$ and $L_2(r)$, then $m \mid h_0$.

Proof. Every common divisor of a and s divides both su and ad , and hence divides $su - ad = h_0$. For the value statement, use the exact Bezout-type identity

$$sL_2(r) - aL_1(r) = s(u + ar) - a(d + sr) = h_0. \tag{6}$$

Every common value divisor therefore divides h_0 . $\square$

Corollary 2.3 (Common prime powers are fixed). *For every prime p , integer $k \geq 1$, and $r \in \mathbb{Z}$,*

$$p^k \mid L_1(r), \quad p^k \mid L_2(r) \implies p^k \mid h_0.$$

In particular, all simultaneous prime divisors belong to the fixed finite set $\{p : p \mid h_0\}$.

Proof. Apply (6). $\square$

Remark 2.4 (What is and is not fixed). The prime divisors of $\gcd(a, s)$ are confined to h_0 , but the prime divisors of nonzero a or s separately need not be. Thus a growing family can have new one-slope primes. If one slope vanishes, primitivity forces the corresponding affine form to be the constant unit 1 or -1 ; that degenerate one-form branch is recorded separately in section 3. None of these primes is automatically a common-root prime.

2.1 Exact content audit

The primitive hypothesis cannot be imposed by silently dividing coefficients. The next proposition records the precise reduction.

Proposition 2.5 (Content factorization). *For an arbitrary affine pair satisfying (2), put*

$$c_1 = \gcd(d, s), \quad c_2 = \gcd(u, a),$$

and write

$$L_1 = c_1 L_1^*, \quad L_2 = c_2 L_2^*.$$

Then L_1^, L_2^* are individually primitive and*

$$c_1 c_2 \mid h_0, \quad \det(L_1^*, L_2^*) = \frac{h_0}{c_1 c_2} \neq 0. \tag{7}$$

If $p^2 \mid c_i$, then $\mu(L_i(r)) = 0$ whenever $L_i(r) > 0$. If c_i is squarefree, then

$$\mu(c_i L_i^*(r)) = \mu(c_i) \mu(L_i^*(r)) \mathbf{1}_{\gcd(c_i, L_i^*(r))=1} \tag{8}$$

on positive values.

Proof. Write

$$d = c_1 d^*, \quad s = c_1 s^*, \quad u = c_2 u^*, \quad a = c_2 a^*.$$

Then

$$h_0 = su - ad = c_1 c_2 (s^* u^* - a^* d^*),$$

which proves (7). The reduced coefficient pairs are primitive by the definition of content.

If $p^2 \mid c_i$, then $p^2 \mid L_i(r)$ for every r , so the Mobius value vanishes. If c_i is squarefree, $\mu(c_i n)$ equals $\mu(c_i)\mu(n)$ when $(c_i, n) = 1$ and vanishes when $(c_i, n) > 1$, proving (8). $\square$

Remark 2.6 (Literal status of content reduction). Because $c_1 c_2 \mid h_0$, all content primes come from a fixed finite set. Nevertheless, (8) introduces an actual local mask, and the reduced determinant is $h_0/(c_1 c_2)$, not h_0 . Verifying that the physical packet is already primitive, or tracking these constants and masks through the outer sum, is an L1 task. The rest of the paper works on the primitive branch and never relabels the reduced determinant as the original one.

3 Exact local root counts

Assume in this section that (L_1, L_2) is individually primitive. For a prime p and $k \geq 1$, define

$$\begin{aligned} \mathcal{R}_{1;p,k} &= \{r \bmod p^k : p^k \mid L_1(r)\}, \\ \mathcal{R}_{2;p,k} &= \{r \bmod p^k : p^k \mid L_2(r)\}, \\ \mathcal{R}_{p,k} &= \mathcal{R}_{1;p,k} \cup \mathcal{R}_{2;p,k}, \quad \nu_{p,k} = \#\mathcal{R}_{p,k}. \end{aligned} \tag{9}$$

Also put

$$\epsilon_{1,p} = \mathbf{1}_{p \nmid s}, \quad \epsilon_{2,p} = \mathbf{1}_{p \nmid a}.$$

Lemma 3.1 (One primitive linear congruence). *For every prime p and $k \geq 1$,*

$$\#\mathcal{R}_{1;p,k} = \epsilon_{1,p}, \quad \#\mathcal{R}_{2;p,k} = \epsilon_{2,p}.$$

More precisely, if $p \nmid s$, then

$$\mathcal{R}_{1;p,k} = \{-ds^{-1} \bmod p^k\},$$

whereas if $p \mid s$, then $\mathcal{R}_{1;p,k}$ is empty. The analogous statement holds with (d, s) replaced by (u, a) .

Proof. If $p \nmid s$, the slope is invertible modulo p^k , so there is exactly one solution. If $p \mid s$, primitivity gives $p \nmid d$, and $d + sr \not\equiv 0 \pmod{p}$ for every r . Therefore there is no solution modulo p^k . The proof for L_2 is identical. $\square$

Lemma 3.2 (Common-root criterion). *For every prime p and $k \geq 1$,*

$$\mathcal{R}_{1;p,k} \cap \mathcal{R}_{2;p,k} \neq \emptyset \iff p \nmid as \text{ and } p^k \mid h_0. \tag{10}$$

When the intersection is nonempty, it consists of one residue class.

Proof. If the two root sets meet, lemma 3.1 gives $p \nmid as$, and corollary 2.3 gives $p^k \mid h_0$.

Conversely, suppose $p \nmid as$ and $p^k \mid h_0$. Let r_1 be the unique root of L_1 modulo p^k . From (6),

$$sL_2(r_1) = h_0 + aL_1(r_1) \equiv 0 \pmod{p^k}.$$

Since s is a unit modulo p^k , $L_2(r_1) \equiv 0 \pmod{p^k}$. Thus the unique roots coincide. $\square$

Theorem 3.3 (Complete prime-power root count). *For every prime p and $k \geq 1$,*

$$\boxed{\nu_{p,k} = \epsilon_{1,p} + \epsilon_{2,p} - \epsilon_{1,p}\epsilon_{2,p}\mathbf{1}_{p^k \mid h_0}.} \tag{11}$$

Equivalently,

<i>slope status at p</i>	<i>determinant status</i>	$\nu_{p,k}$
$p \nmid as$	$p^k \nmid h_0$	2
$p \nmid as$	$p^k \mid h_0$	1
$p \mid s, p \nmid a$	<i>arbitrary</i>	1
$p \mid a, p \nmid s$	<i>arbitrary</i>	1
$p \mid \gcd(a, s)$	<i>necessarily $p \mid h_0$</i>	0.

Proof. By inclusion–exclusion,

$$\nu_{p,k} = \#\mathcal{R}_{1;p,k} + \#\mathcal{R}_{2;p,k} - \#(\mathcal{R}_{1;p,k} \cap \mathcal{R}_{2;p,k}).$$

Apply [lemmas 3.1](#) and [3.2](#). The last row also uses [\(4\)](#). □

Corollary 3.4 (The p versus p^2 distinction). *Suppose $p \nmid as$.*

- (i) *The roots of L_1, L_2 coincide modulo p if and only if $p \mid h_0$.*
- (ii) *They coincide modulo p^2 if and only if $p^2 \mid h_0$.*
- (iii) *If $p \mid h_0$ but $p^2 \nmid h_0$, the two roots coincide modulo p but are distinct modulo p^2 .*

Proof. Apply [lemma 3.2](#) with $k = 1, 2$. □

3.1 A careful exceptional-prime taxonomy

Remark 3.5 (The zero-slope unit branch). If $as = 0$, then $h_0 \neq 0$ forces exactly one slope to vanish. Primitivity forces the corresponding intercept to be 1 or -1 , so that form has no root modulo any prime power and contributes no Mobius or squarefree obstruction on a positive interval. The formula [\(11\)](#) remains valid and reduces to the one-form root count. Because every prime divides the integer 0, finite exceptional-set language for “primes dividing the zero slope” would be misleading. The taxonomy below therefore concerns the two-nonconstant branch $as \neq 0$.

Definition 3.6 (Joint and slope-exceptional primes). For a fixed primitive pair with $as \neq 0$, define

$$\mathcal{E}_{\text{joint}} = \{p : p \mid h_0\}, \quad \mathcal{E}_{\text{slope}} = \{p : p \mid as\}.$$

The generic set is the complement of $\mathcal{E}_{\text{joint}} \cup \mathcal{E}_{\text{slope}}$.

Proposition 3.7 (Exceptional-prime audit). *On the branch $as \neq 0$, the following statements hold.*

- (a) *For $p \nmid ash_0$, one has $\nu_{p,k} = 2$ for every $k \geq 1$.*
- (b) *Every common-value or common-root prime lies in the fixed set $\mathcal{E}_{\text{joint}}$.*
- (c) *If $p \mid as$ but $p \nmid h_0$, then p divides exactly one slope and $\nu_{p,k} = 1$ for every $k \geq 1$.*
- (d) *If $p \mid \gcd(a, s)$, then $p \mid h_0$ and neither form has a root modulo p .*

Proof. Part (a) follows from [\(11\)](#). Part (b) is [corollary 2.3](#). For (c), if p divided both slopes then [\(4\)](#) would give $p \mid h_0$; the root count is then the third or fourth row of the table in [theorem 3.3](#). Part (d) follows from [\(4\)](#) and primitivity. □

Remark 3.8 (Fixed versus moving exceptions). The joint set is fixed when h_0 is fixed. The one-slope set may grow with X , so it is inaccurate to say that *all* primes at which the root count differs from two divide h_0 . On the other hand, a one-slope prime removes a forbidden root: it raises, rather than lowers, the squarefree local factor relative to the generic factor $1 - 2/p^2$.

4 Squarefree local support

For r with $L_1(r)L_2(r) \neq 0$, define

$$\mathcal{Q}(r) = \mu^2(|L_1(r)|)\mu^2(|L_2(r)|). \quad (12)$$

At an integral zero of either form, set $\mathcal{Q}(r) = 0$. This convention changes at most two values and is convenient for density statements.

4.1 Finite local support

For $z \geq 2$, let

$$P(z) = \prod_{p \leq z} p^2$$

and define the truncated local indicator

$$\mathcal{Q}_z(r) = \prod_{p \leq z} \mathbf{1}_{r \bmod p^2 \notin \mathcal{R}_{p,2}}. \quad (13)$$

Theorem 4.1 (Exact finite local density). *The function $\mathcal{Q}_z$ is periodic modulo $P(z)$, and*

$$\frac{1}{P(z)} \sum_{r \bmod P(z)} \mathcal{Q}_z(r) = \prod_{p \leq z} \left(1 - \frac{\nu_{p,2}}{p^2}\right). \quad (14)$$

Every factor on the right is positive.

Proof. For each $p \leq z$, exactly $p^2 - \nu_{p,2}$ residue classes modulo p^2 are allowed. The moduli p^2 are pairwise coprime, so the Chinese remainder theorem multiplies the allowed counts. Finally, $\nu_{p,2} \leq 2 < p^2$, including at $p = 2$. $\square$

Corollary 4.2 (No squarefree local annihilation). *For every finite set of primes, there is a residue class satisfying $p^2 \nmid L_i(r)$ for $i = 1, 2$ and every prime in the set. Quantitatively,*

$$\prod_{p \leq z} \left(1 - \frac{\nu_{p,2}}{p^2}\right) \geq \prod_{p \leq z} \left(1 - \frac{2}{p^2}\right) > 0. \quad (15)$$

Proof. Use $\nu_{p,2} \leq 2$ in (14). $\square$

4.2 The Euler product and fixed-form density

Definition 4.3 (Squarefree singular factor). For a fixed primitive affine pair, put

$$\mathfrak{S}_{\text{sf}}(L_1, L_2) = \prod_p \left(1 - \frac{\nu_{p,2}}{p^2}\right). \quad (16)$$

Proposition 4.4 (Absolute convergence and uniform positivity). *The product (16) converges to a positive number and satisfies*

$$1 \geq \mathfrak{S}_{\text{sf}}(L_1, L_2) \geq \prod_p \left(1 - \frac{2}{p^2}\right) > 0. \quad (17)$$

Proof. Since $0 \leq \nu_{p,2} \leq 2$ and $\sum_p p^{-2} < \infty$, the logarithm of the Euler product converges absolutely after finitely many initial factors. Every factor is positive. The lower bound follows termwise. $\square$

Theorem 4.5 (Simultaneous squarefree density for a fixed pair). *For every fixed primitive affine pair with $h_0 \neq 0$,*

$$\lim_{R \rightarrow \infty} \frac{1}{R} \sum_{1 \leq r \leq R} \mathcal{Q}(r) = \mathfrak{S}_{\text{sf}}(L_1, L_2). \quad (18)$$

Proof. Fix z . Periodicity and [theorem 4.1](#) give

$$\lim_{R \rightarrow \infty} \frac{1}{R} \sum_{r \leq R} \mathcal{Q}_z(r) = \prod_{p \leq z} \left(1 - \frac{\nu_{p,2}}{p^2}\right).$$

It remains to remove primes $p > z$.

Let

$$B_R = \max_{1 \leq r \leq R} \max\{|L_1(r)|, |L_2(r)|\}.$$

For the fixed affine pair, $B_R \ll_{L_1, L_2} R$. Away from the at most two integral zeros, a discrepancy $\mathcal{Q}_z(r) \neq \mathcal{Q}(r)$ requires

$$p^2 \mid L_i(r)$$

for some $i \in \{1, 2\}$ and prime $z < p \leq \sqrt{B_R}$. If p divides the relevant slope, there is no root by [lemma 3.1](#); otherwise there is one residue class modulo p^2 . Hence the number of discrepancies is at most

$$2R \sum_{p > z} \frac{1}{p^2} + 2\pi(\sqrt{B_R}) + O(1) \ll \frac{R}{z} + \sqrt{R} + O(1).$$

After division by R , first let $R \rightarrow \infty$, then $z \rightarrow \infty$. The finite products converge to [\(16\)](#), proving the theorem. $\square$

Remark 4.6 (What the density proves). The theorem proves that a *fixed* primitive pair has positive simultaneous squarefree density. It does not prove cancellation of Mobius or Liouville signs. It also does not give a uniform translated-short-interval theorem for coefficients depending on X .

4.3 Why growing physical fibers need a separate theorem

Proposition 4.7 (Elementary tail bound on one finite interval). *Let $I \subset \mathbb{Z}$ be an interval of N consecutive integers, and put*

$$B = \max_{r \in I} \max\{|L_1(r)|, |L_2(r)|\}.$$

Apart from integral zeros of the forms,

$$\#\{r \in I : \mathcal{Q}_z(r) \neq \mathcal{Q}(r)\} \ll \frac{N}{z} + \sqrt{B}. \quad (19)$$

The implied constant is absolute.

Proof. Repeat the union bound in the proof of [theorem 4.5](#). A single soluble class modulo p^2 meets I at most $N/p^2 + 1$ times, and only primes $p \leq \sqrt{B}$ can occur. $\square$

Remark 4.8 (A limitation, not a counterexample). If the slopes and interval origins grow, then B may be much larger than N^2 , and [\(19\)](#) gives no useful relative error. This does not prove that squarefree mass is small; it proves only that the fixed-form density argument is not a uniform physical theorem. Likewise, the exact periodic density in [\(14\)](#) need not be visible inside an interval much shorter than $P(z)$.

5 Exact squarefree-support and parity splitting

Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function. For every $n \in \mathbb{N}$,

$$\boxed{\mu(n) = \mu^2(n)\lambda(n)}. \quad (20)$$

Indeed, both sides vanish when n is not squarefree, while on a squarefree integer both equal $(-1)^{\omega(n)}$.

Theorem 5.1 (Exact two-form split). *Let $I \subset \mathbb{Z}$ be a finite interval on which $L_1(r), L_2(r) > 0$, and let $W : I \rightarrow \mathbb{C}$ be arbitrary. Then*

$$\boxed{\sum_{r \in I} W(r) \mu(L_1(r)) \mu(L_2(r)) = \sum_{r \in I} W(r) \mathcal{Q}(r) \lambda(L_1(r)) \lambda(L_2(r))}. \quad (21)$$

The factor $\mathcal{Q}$ is determined entirely by the square-power local sets $\mathcal{R}_{p,2}$; the remaining factor is a two-point Liouville sign.

Proof. Apply (20) to both forms, multiply, and sum. $\square$

Remark 5.2 (Support is not sign cancellation). The positive density and exact root counts of [sections 3](#) and [4](#) concern only $\mathcal{Q}$. They do not estimate the signed sum on the right of (21). Replacing the Liouville product by the constant sign 1 preserves every local-support statement and produces no cancellation for nonnegative W . This is a logical separation, not a model for the actual Liouville function.

5.1 Exact square-divisor expansion

The identity

$$\mu^2(n) = \sum_{q^2 | n} \mu(q) \quad (22)$$

turns the local support into congruence classes. The determinant gives a particularly sharp compatibility condition.

Theorem 5.3 (Primitive square-divisor compatibility). *Let q_1, q_2 be positive squarefree integers and put $g = \gcd(q_1, q_2)$. For a primitive pair (L_1, L_2) , the system*

$$q_1^2 \mid L_1(r), \quad q_2^2 \mid L_2(r) \quad (23)$$

has an integral solution r if and only if

$$\boxed{\gcd(q_1, s) = 1, \quad \gcd(q_2, a) = 1, \quad g^2 \mid h_0.} \quad (24)$$

When soluble, (23) defines exactly one residue class modulo

$$\text{lcm}(q_1^2, q_2^2) = \frac{q_1^2 q_2^2}{g^2}. \quad (25)$$

Proof. The first congruence in (23) is soluble if and only if $(q_1, s) = 1$. To see this prime by prime, if $p \mid (q_1, s)$, primitivity gives $p \nmid d$, so there is no root even modulo p ; if $(q_1, s) = 1$, there is one root modulo q_1^2 . The second congruence is analogous.

Under these two coprimality conditions, the individual congruences specify one class modulo q_1^2 and one class modulo q_2^2 . The generalized Chinese remainder theorem says they are compatible exactly when their reductions agree modulo $\gcd(q_1^2, q_2^2) = g^2$. Since q_1, q_2 are squarefree, this can be checked at every prime $p \mid g$. Both slopes are then units modulo p , and [lemma 3.2](#) with $k = 2$ says that the roots agree modulo p^2 exactly when $p^2 \mid h_0$. Thus compatibility is equivalent to $g^2 \mid h_0$. The unique class modulo $\text{lcm}(q_1^2, q_2^2)$ is the standard conclusion of the generalized Chinese remainder theorem. $\square$

Corollary 5.4 (Exact weighted divisor expansion). *Under the hypotheses of [theorem 5.1](#),*

$$\begin{aligned} & \sum_{r \in I} W(r) \mu(L_1(r)) \mu(L_2(r)) \\ &= \sum_{\substack{q_1, q_2 \geq 1 \\ (q_1, s) = (q_2, a) = 1 \\ \gcd(q_1, q_2)^2 \mid h_0}} \mu(q_1) \mu(q_2) \sum_{\substack{r \in I \\ q_1^2 \mid L_1(r) \\ q_2^2 \mid L_2(r)}} W(r) \lambda(L_1(r)) \lambda(L_2(r)). \end{aligned} \quad (26)$$

The outer sum is finite: only $q_i^2 \leq \max_{r \in I} L_i(r)$ can contribute. Each nonempty inner sum is over one residue class modulo $\text{lcm}(q_1^2, q_2^2)$.

Proof. Insert (22) twice into the right-hand side of (21), interchange the finite sums, and apply theorem 5.3. Terms with nonsquarefree q_i vanish because $\mu(q_i) = 0$. $\square$

Remark 5.5 (What the divisor expansion exposes). The overlap $\gcd(q_1, q_2)$ is supported only on prime squares dividing h_0 , a fixed finite resource. However, (26) still contains Liouville correlations on progressions whose moduli, endpoints, and coefficients can grow. The exact local resolution therefore organizes the parity problem; it does not eliminate it.

5.2 A fixed-form consequence and its boundary

Corollary 5.6 (Unsigned support mass). *For a fixed primitive pair,*

$$\sum_{r \leq R} \mathcal{Q}(r) = \mathfrak{S}_{\text{sf}}(L_1, L_2)R + o(R).$$

Consequently,

$$\left| \sum_{\substack{1 \leq r \leq R \\ L_1(r) \overline{L_2(r)} \neq 0}} \mu(|L_1(r)|) \mu(|L_2(r)|) \right| \leq \mathfrak{S}_{\text{sf}}(L_1, L_2)R + o(R).$$

Proof. The first statement is theorem 4.5. For every nonzero integer n , the identity $\mu(|n|) = \mu^2(|n|)\lambda(|n|)$ holds. Apply it to both forms and use $|\lambda| = 1$; the at most two integral zeros have already been excluded. $\square$

Remark 5.7 (No nontrivial signed saving). The last upper bound is the support-mass bound and is generally of order R . It is not a Chowla estimate and supplies no $o(R)$ cancellation.

6 The literal growing fixed- h_0 correlation

We now index the forms by a scale X and an outer key $\theta \in \Theta_X$:

$$L_{1,\theta,X}(r) = d_{\theta,X} + s_{\theta,X}r, \quad L_{2,\theta,X}(r) = u_{\theta,X} + a_{\theta,X}r. \quad (27)$$

The literal branch requires

$$s_{\theta,X}u_{\theta,X} - a_{\theta,X}d_{\theta,X} = h_0 \neq 0 \quad (28)$$

with the same prescribed h_0 for every active key. It also requires individual primitivity, or the exact content audit of proposition 2.5.

Let $I_{\theta,X}$ be the actual translated interval on which both forms are positive. Combine the physical smooth weight, coprimality and residue masks, joint mask, polarization, and every other local multiplier into one literal coefficient

$$W_{\theta,X} : I_{\theta,X} \rightarrow \mathbb{C}.$$

No assumption of positivity or translation invariance is made.

Definition 6.1 (Literal parity correlation). Define

$$C_{\theta,X} = \sum_{r \in I_{\theta,X}} W_{\theta,X}(r) \mu(L_{1,\theta,X}(r)) \mu(L_{2,\theta,X}(r)), \quad (29)$$

$$\mathcal{Q}_{\theta,X}(r) = \mu^2(L_{1,\theta,X}(r)) \mu^2(L_{2,\theta,X}(r)). \quad (30)$$

Theorem 6.2 (Exact literal local-parity split). *For every active key,*

$$C_{\theta,X} = \sum_{r \in I_{\theta,X}} W_{\theta,X}(r) \mathcal{Q}_{\theta,X}(r) \lambda(L_{1,\theta,X}(r)) \lambda(L_{2,\theta,X}(r)). \quad (31)$$

At each fixed X, θ , the local sets and root counts of $\mathcal{Q}_{\theta,X}$ are given exactly by theorem 3.3. In particular, all simultaneous root primes divide h_0 .

Proof. Apply theorem 5.1 to the literal interval and weight. The local classification follows from theorems 2.2 and 3.3. $\square$

6.1 The actual unresolved estimate

In the determinant packet introduced in TPC-32 [4], suppose the complete raw distinguished coefficient has an exact outer reassembly

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} C_{\theta,X} + E_{\text{content},X}, \quad (32)$$

where all native keys and both polarizations occur once, the $c_{\theta,X}$ are the literal outer coefficients, and $E_{\text{content},X}$ contains only explicitly audited content or bounded-fiber terms.

Definition 6.3 (Growing fixed-determinant parity gate). The *literal growing fixed- h_0 weighted parity gate* is the problem of obtaining the required saving for

$$\sum_{\theta \in \Theta_X} c_{\theta,X} \sum_{r \in I_{\theta,X}} W_{\theta,X}(r) \mathcal{Q}_{\theta,X}(r) \lambda(L_{1,\theta,X}(r)) \lambda(L_{2,\theta,X}(r)), \quad (33)$$

uniformly in the actual active family and with every physical loss and completion term included.

Equation (33), not the Euler product, is the remaining arithmetic object. A strong fiberwise theorem sufficient for many partial-summation or dyadic-reassembly arguments would control every relevant prefix $I_{\theta,X}(M)$:

$$\sup_{\substack{\theta \in \Theta_X \\ M \text{ an active prefix}}} \left| \sum_{r \in I_{\theta,X}(M)} W_{\theta,X}(r) \mathcal{Q}_{\theta,X}(r) \lambda(L_{1,\theta,X}(r)) \lambda(L_{2,\theta,X}(r)) \right| \leq X^{-\eta+o(1)} \mathcal{M}_{\theta,X}(M), \quad (34)$$

where

$$\mathcal{M}_{\theta,X}(M) = \sum_{r \in I_{\theta,X}(M)} |W_{\theta,X}(r)| \mathcal{Q}_{\theta,X}(r).$$

Here $\eta > 0$ is only a schematic desired saving. Equation (34) is not asserted to be necessary: cancellation may also occur in the exact outer sum (32). The minimal requirement is the calibrated bound on that complete sum.

Proposition 6.4 (Why fixed-form local information is insufficient). *The results proved in sections 2 to 5 do not imply either (34) or a power saving for (33).*

Proof. Those results determine the zero set of $\mathcal{Q}_{\theta,X}$, its finite periodic densities, and the exact placement of square divisors. They contain no estimate for the signs $\lambda(L_{1,\theta,X})\lambda(L_{2,\theta,X})$. Even the fixed-form density theorem is unsigned and asymptotic in an interval whose length tends to infinity with fixed coefficients. It supplies neither uniformity in the growing coefficients nor control of translated short prefixes and literal weights. Therefore the stated signed conclusions do not follow. $\square$

6.2 Comparison with available correlation theorems

Tao's logarithmically averaged two-point theorem fixes the affine coefficients before the main scale tends to infinity [3]. The averaged Chowla theorem of Matomäki et al. [1] averages over a shift box whose side length $H = H(X)$ tends to infinity with the scale. Neither quantifier pattern gives (33): the physical problem prescribes one determinant, lets the slopes and intercepts grow, retains ordinary local weights and masks, and then performs a specific outer sum. Moreover, replacing a weight $1/(r_0 + r)$ by $1/r$ after translating a fiber requires a uniform partial-summation theorem; it is not a formal relabeling. These distinctions are exactly the Tauberian and literal-coefficient gaps isolated in TPC-87 [8].

Remark 6.5 (No specialization). The determinant in (28) remains an arbitrary prescribed nonzero integer. None of the local identities or conditional targets selects $h_0 = 2$, produces prime values, or proves a fixed-shift Hardy–Littlewood asymptotic.

7 Independent compatibility and physical-loss ledgers

The local split introduces no automatic fixed-power saving. To prevent local density, sign cancellation, determinant energy, and physical reassembly from being charged against one another, we keep their ledgers explicit.

7.1 Determinant compatibility

At the inherited critical scales and with the compatibility bookkeeping of TPC-81 and TPC-82 [6, 7], write

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}. \quad (35)$$

Suppose a future theorem proves the complete literal zero-mode bound

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2 \quad (36)$$

and an independent theorem proves the determinant-energy lower bound

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2. \quad (37)$$

These are hypotheses; neither is proved here.

Proposition 7.1 (Compatibility exponent). *Under (35)–(37),*

$$\frac{|Z_X|^2}{D_X} \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}. \quad (38)$$

Thus the recorded bounds preserve the baseline exponent whenever

$$\boxed{\lambda_D \leq 2\eta_Z.} \quad (39)$$

Proof. Squaring (36) and dividing by (37) gives

$$\frac{|Z_X|^2}{D_X} \leq X^{\lambda_D-2\eta_Z+o(1)} \frac{Q_X}{J_X^2}.$$

By (35),

$$\frac{Q_X}{J_X^2} = X^{1/400+o(1)}.$$

This proves (38) and (39). □

Remark 7.2 (Local density is not η_Z). The factor $\mathfrak{S}_{\text{sf}}(L_1, L_2)$ measures unsigned squarefree support mass. Its positivity or boundedness away from zero is not a signed zero-mode saving and cannot be entered as a positive contribution to η_Z . Any actual η_Z must come from cancellation in the complete literal sum (33), possibly together with a separately proved outer-reassembly mechanism.

7.2 The independent physical-loss rule

Definition 7.3 (Physical loss). Let Λ_{phys} be the sum of fixed-power losses actually incurred by coefficient growth, interval localization, squarefree-tail localization, masks, short-fiber treatment, content reduction, both polarizations, outer reassembly, dictionary transfer, and downstream calibrated drift, after removing any loss already included in $\lambda_D - 2\eta_Z$.

The inherited endpoint rule is

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}.} \quad (40)$$

Equality or excess is a stop condition.

Proposition 7.4 (The ledgers are independent). *None of the following conversions is licensed by the results of this paper:*

- (i) replacing a sign-cancellation gain η_Z by the positive local factor $\mathfrak{S}_{\text{sf}}$;
- (ii) subtracting a squarefree-density constant from Λ_{phys} ;
- (iii) treating an unproved uniform support-tail estimate as determinant energy;
- (iv) adding (39) and (40) into one budget;
- (v) charging a loss already present in $\lambda_D - 2\eta_Z$ a second time in Λ_{phys} .

Proof. The compatibility ledger compares a signed zero mode with an ℓ^2 determinant energy. The local factor describes unsigned residue support. The physical ledger records losses created by literal localization and reassembly. No theorem in this paper identifies these quantities, so no conversion follows. $\square$

Ledger	Required comparison	Status in this paper
Local support	$\nu_{p,2}$ and $\mathfrak{S}_{\text{sf}}$ describe squarefree residue mass	Exact L0; no sign saving
Determinant compatibility	$\lambda_D \leq 2\eta_Z$	Conditional L1 bookkeeping; neither analytic input proved
Physical reassembly	$\Lambda_{\text{phys}} < 1/400$	Independent inherited stop rule

Table 1: Unsigned local mass, signed cancellation, determinant energy, and physical loss have different inputs.

Remark 7.5 (Where a local-uniformity cost belongs). If a future proof truncates the square-divisor expansion and pays a fixed-power cost to make (19) uniform over the physical family, that cost belongs in Λ_{phys} unless it has already been explicitly incorporated into η_Z or λ_D . The proof must state which ledger contains it.

8 Claim boundary and conclusion

8.1 What is proved

Theorem 8.1 (Audited primitive affine split). *Let*

$$L_1(r) = d + sr, \quad L_2(r) = u + ar$$

be individually primitive integral affine forms with $su - ad = h_0 \neq 0$. Then:

$$(i) \quad \gcd(a, s) \mid h_0, \quad \gcd(L_1(r), L_2(r)) \mid h_0;$$

(ii) *for every prime p and $k \geq 1$,*

$$\nu_{p,k} = \mathbf{1}_{p \nmid s} + \mathbf{1}_{p \nmid a} - \mathbf{1}_{p \nmid as} \mathbf{1}_{p^k \mid h_0};$$

(iii) *the simultaneous squarefree local product is positive and uniformly bounded below:*

$$\mathfrak{S}_{\text{sf}}(L_1, L_2) = \prod_p \left(1 - \frac{\nu_{p,2}}{p^2}\right) \geq \prod_p \left(1 - \frac{2}{p^2}\right) > 0;$$

- (iv) for fixed coefficients, $\mathfrak{S}_{\text{sf}}(L_1, L_2)$ is the natural density of r for which both values are squarefree;
(v) for squarefree q_1, q_2 , the simultaneous square-divisor system is soluble exactly under

$$(q_1, s) = (q_2, a) = 1, \quad \gcd(q_1, q_2)^2 \mid h_0,$$

and then gives one class modulo $\text{lcm}(q_1^2, q_2^2)$;

- (vi) on every positive finite interval and for every complex weight W ,

$$\sum W \mu(L_1) \mu(L_2) = \sum W \mu^2(L_1) \mu^2(L_2) \lambda(L_1) \lambda(L_2).$$

Proof. Combine [theorems 2.2, 3.3, 4.5, 5.1](#) and [5.3](#) and [proposition 4.4](#). □

All statements in [theorem 8.1](#), including the fixed-form density theorem, are L0. They apply to the abstract affine pair exactly as stated.

8.2 What requires a literal L1 verification

To use the split in the physical packet, one must verify all of the following:

- (a) the displayed coefficients are the literal growing coefficients and satisfy the same prescribed nonzero h_0 ;
- (b) each pair is primitive, or the exact content constants, changed determinant, finite coprimality masks, and identically zero branches of [proposition 2.5](#) are retained;
- (c) the interval origin, positivity range, all residue, coprimality, smooth, and joint masks, and all logarithmic or ordinary weights are unchanged;
- (d) both polarizations and every native outer key are present;
- (e) short or bounded-length fibers receive a separate exact or outer estimate;
- (f) every prefix required by partial summation or dyadic recursion is controlled;
- (g) [\(32\)](#) is the complete raw physical coefficient under one global post-aggregation normalization;
- (h) the physical-loss ledger satisfies $\Lambda_{\text{phys}} < 1/400$.

Only after these checks does [\(31\)](#) become a literal L1 certificate rather than an abstract identity.

8.3 What is not proved

This paper does not prove:

- (a) a uniform squarefree-density asymptotic on the actual growing, translated, possibly short physical fibers;
- (b) cancellation of $\lambda(L_{1,\theta,X})\lambda(L_{2,\theta,X})$ on the literal squarefree support;
- (c) the prefix estimate [\(34\)](#);
- (d) a power saving for the complete outer correlation [\(33\)](#);
- (e) uniformity of a fixed-form logarithmic theorem in the growing physical coefficients;
- (f) selection of the prescribed h_0 from an averaged-shift theorem;
- (g) a determinant-energy lower bound for D_X ;
- (h) either conditional hypothesis [\(36\)](#) or [\(37\)](#);
- (i) a fixed- h_0 Hardy–Littlewood asymptotic, a breach of the sieve parity barrier, or a prime-pair lower bound.

These missing signed estimates are L2, not consequences of the local classification.

8.4 Conclusion and next gate

The primitive affine determinant resolves the local geometry more completely than a generic “finite bad primes” slogan. It proves two different facts:

common value primes divide h_0 ,

while

one-slope primes may move but delete one local root.

At square level, the exact formula

$$\nu_{p,2} = \mathbf{1}_{p \nmid s} + \mathbf{1}_{p \nmid a} - \mathbf{1}_{p \nmid as} \mathbf{1}_{p^2 \mid h_0}$$

shows that the squarefree support has no local annihilation. The identity $\mu = \mu^2 \lambda$ then places every remaining sign issue in the explicit correlation (33).

The next positive gate is therefore not another local-density calculation. It is a theorem that gains cancellation from the Liouville product while retaining the literal growing forms, fixed nonzero determinant, support, origins, weights, masks, prefixes, and outer reassembly. A viable weaker step could control one literal residual block or a certified range of square-divisor moduli, provided it is labeled at its actual proof level and its loss is entered in the correct ledger. A proof that such a route necessarily loses 1/400 or more would be a valid negative endpoint.

Nothing here specializes h_0 to 2, proves that the affine values are prime, gives a prime-pair lower bound, or proves the twin-prime conjecture. The result is an exact local-obstruction split and a sharper statement of the parity gate.

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